ORIENTATION-BIT GEOMETRY OF THE MÖBIUS PATH: REFLECTION FIBRES, DIVISOR–SHIFT BOUNDARY COCYCLES, AND AN OBSERVABILITY ROUTE TO TWO-POINT CANCELLATION

Date: Version 6.

Abstract. This notebook develops an operator framework for the two-point correlation $C_N(h) = \sum_{n \le N-h} \mu(n)\mu(n+h)$ built on the orientation bits of the Möbius path and on the divisor operator.

Version 3 adds an analytic layer to the combinatorial one. The central new object is the difference $(U_h - I)D$ rather than the commutator or the conjugation: each divisor column collapses to the difference of two residue classes, vanishes identically when the divisor divides h, and has interval sums bounded by one. A purely algebraic Haar basis over an arbitrary binary tree then annihilates the volumetric CRT term exactly, leaving only boundary cocycles which satisfy a parent–children additivity law. No doubling, no comparable lengths and no regularity of the squarefree mask are required at any point.

Four further results are proved. An exact observability criterion: $|x^{\mathsf{T}} A x| \le \kappa \|Lx\|^2$ holds for some finite κ if and only if $\ker L \subseteq \ker A$, with the optimal constant and the exact operator-norm dual given in closed form. A no-go: uniform observability of the squarefree shift by a sublinear set of divisor rows is impossible, because above $N/2$ each column is seen only by its own row and shifted squarefree pairs have positive density. An extension of the blindness theorem to kernels that are constant on each half of an excursion, under which the cross-excursion interaction factors through the single channel $s_j H_j$ and the shift kernel is shown not to belong to the extended class. And a geometric proposition: the hypothesis $u_j \ge m_j/2$ needed for the central divisor matrix to be triangular with nonvanishing diagonal fails for at most $\lfloor \log_2 N \rfloor$ excursions, so that the Möbius line is isolated up to a kernel of logarithmic dimension.

The results of Versions 1 and 2 are retained: the reflection fibre of the quadratic profile, the exactly isotropic null, the blindness theorem for kernels depending on $\max(a, b)$, and the criterion separating tautological from informative sine-kernel content. Negative results, including the collapse of every multiplicative diagonal weight onto $1/\zeta(1 + s)$, are recorded in full. No numerical computation is performed.

Version 4 audits the target objective of Version 3 and finds it self-defeating as stated. An exact identity expresses the L^2 defect of any leaf projection in the squarefree space as an explicit combination of within-leaf correlations of μ along consecutive squarefree integers. The capture condition of the adaptive boundary objective therefore demands maximally positive correlations on every nontrivial leaf, the opposite of the cancellation it is meant to produce, while long leaves are excluded unconditionally by short-interval cancellation. The objective is marked superseded. The engine common to both objectives, the annihilation $D\mu = e_1$, survives in the central-row objective, which becomes the live target. The conditioning of the two-point expansion on central data and the redundancy of quarter and octave evaluations are also recorded.

Version 5 incorporates an external audit, verified line by line. It corrects the pushed form to $\widehat{G} = F^{\mathsf{T}} G F$ with incremental factors, fixes a single orientation basis, states the conditioning on $M(N)$ that freezes the terminal block, corrects the parity statement for single reflections, and withdraws two overstatements (a semicircular density inferred from an edge bound, and a general dichotomy for non-multiplicative weights). It withdraws the orthogonal-to-unitary crossover conjecture as stated: a Hermitian deformation supported on a forest, or obtained by unitary transport, is isospectral, so a crossover needs flux around cycles. It adds four exact results: the energy kernel is Montgomery's weight localised at $\alpha = 0$ with tail $\Pr(|\alpha| > A) = T^{-2A}$; the Hermitian collision constant differs from the Möbius energy constant by an exact fourth-order term; the form $\mathcal{A}_0'$ is the discrete Lévy area of the planar path (M_Ω, L), neither a product nor a fixed-shift correlation; and the central divisor rows cut the reflection fibre from 2^Z orientations to at most $2N$, with at most $\lfloor \log_2 N \rfloor + 1$ further divisor rows selecting the Möbius orientation uniquely.

Version 6 executes the uncompressed, irregular route analytically and closes it as a family. The interior divisor–shift operator $B_h^\circ = R_{N-h}(U_h - I)D$ has maximal rank $N - h$: the vanishing of the columns $d \mid h$ removes only $\tau(h)$ columns. Each nonzero column is the exact discrete derivative of a periodic block of h mod d consecutive residues, which yields an optimal variation inequality on arbitrary intervals, but the Haar energy of each column is independent of the tree. The quantitative no-go is sharpened to $|\Lambda| \ge \frac{\mathfrak{S}_h}{4} N + o_h(N)$ divisor rows for any uniform observability with error below $1/2$. The shift does not descend to valuation-indexed local factors. The maximal class of kernels blind to zero-sum blocks is exactly the additively separable class, the shift lies at macroscopic distance from it, and no separable approximation of sublinear rank is close in operator norm. What remains open inside this family is a collective signed cancellation of CRT boundary errors with Möbius-linked coefficients, which already requires control of the parity of large primes.

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Remark 0.1. This notebook is exploratory. Rigorous statements appear inside theorem, proposition, lemma or corollary environments with proofs. Results established in separate work are collected in Section 2 and stated without proof. Everything else is labelled heuristic, conjectural or objective. No numerical computation is performed; the empirical values in Remark 15.11 are inputs from separate work.

Warning 0.2. Bibliographic metadata in Section 19 has not been verified against primary sources. DOIs are omitted where not confirmed.

1. Introduction

1.1. **Objects.** Let μ be the Möbius function, $q(n) = \mu(n)^2$,

$$M(x) = \sum_{n \le x} \mu(n), \qquad C_N(h) = \sum_{n \le N-h} \mu(n)\mu(n+h). \tag{1.1}$$

Let D be the divisor operator $(Dx)(r) = \sum_{d|r} x(d)$ and U_h the truncated shift $(U_h x)(n) = x(n+h)$, with symmetric part $A_h = (U_h + U_h^*)/2$, so that

$$C_N(h) = \mu^{\mathsf{T}} A_h \mu. \tag{1.2}$$

1.2. **What is proved in this notebook.**

(i) *Reflection fibre* (Prop. 5.2): the fibre of the quadratic profile is exactly an orbit of $\mathcal{R}_N \cong (\mathbb{Z}/2)^{Z(N)}$.
(ii) *Exact isotropy* (Prop. 5.4): $\mathbb{E}_\nu[ss^{\mathsf{T}}] = I_Z$ with no correction term, and closed-form variance.
(iii) *Blindness* (Thm. 7.2): every cross-excursion contribution to a $\max(a,b)$-kernel vanishes identically.
(iv) *Extended blindness* (Thm. 7.5): for kernels constant on each half of an excursion the cross-excursion interaction factors through the rank-one channel $s_j H_j$; the shift kernel is shown not to lie in this class.
(v) *Divisor–shift boundary lemma* (Lem. 8.2): $((U_h - I)D)_{n,d} = \mathbf{1}[d \mid n+h] - \mathbf{1}[d \mid n]$, identically zero for $d \mid h$, supported on two residue classes otherwise, with interval sums bounded by 1.
(vi) *Exact CRT pullback* (Lem. 8.8): $(D^{\mathsf{T}} U_h D)_{d,e}$ vanishes unless $(d,e) \mid h$ and otherwise equals $(N-h)/[d,e]$ up to an error of modulus at most 1.
(vii) *Algebraic Haar and exact bulk vanishing* (Lem. 9.1, Thm. 9.3): on an arbitrary binary tree the volumetric CRT term is annihilated exactly and only a boundary cocycle survives, additive over children.
(viii) *Observability criterion* (Thm. 10.2): $|x^{\mathsf{T}} A x| \le \kappa \|Lx\|^2$ for some $\kappa < \infty$ iff $\ker L \subseteq \ker A$; optimal constant and operator-norm dual in closed form.
(ix) *Sublinear no-go* (Thm. 10.4): no nontrivial observability of the squarefree shift by $o(N)$ divisor rows.
(x) *Logarithmic exceptional set* (Prop. 11.6): the geometric hypothesis making the central divisor matrix triangular with nonvanishing diagonal fails for at most $\lfloor \log_2 N \rfloor$ excursions.
(xi) *Central dichotomy* (Lem. 11.1): every linear reflection-odd functional of an excursion either vanishes on the whole fibre or reads the orientation bit directly.
(xii) *Leaf-energy identity and squeeze* (Prop. 17.4, Cor. 17.6): the L^2 defect of a leaf projection equals $\sum_I \big(k_I - 1 - \frac{2}{k_I} \sum_{i<i' \in I} \mu(a_i)\mu(a_{i'})\big)$, so L^2 capture of μ by a subspace independent of its signs forces either essentially the whole squarefree space or maximal positive within-leaf correlation.
(xiii) *Gauge obstruction* (Prop. 16.4): Hermitian deformations on forests, and unitary transports, are isospectral.
(xiv) *Collision versus energy constant* (Prop. 16.7): exact fourth-order difference.
(xv) *Lévy-area form* (Prop. 16.10): $\mathcal{A}_0'(N)$ is the discrete signed area of (M_Ω, L).
(xvi) *Compressed selection* (Cor. 11.8, Cor. 11.9): at most $2N$ admissible orientations, and unique selection after $\lfloor \log_2 N \rfloor + 1$ further divisor rows.
(xvii) *Interior operator* (Thm. 8.4, Prop. 8.7, Prop. 9.5): $\operatorname{rank} B_h^\circ = N - h$; columns are discrete derivatives of periodic residue blocks; Haar energy is independent of the tree.

(xviii) *Blind class* (Thm. 7.7): kernels blind to zero-sum blocks are exactly $\alpha(a)+\beta(b)$; the shift is at macroscopic distance.

1.3. **Inputs and the methodological point.** Results established in separate work are listed with status in Section 2; the two used actively are the tensor factorisation (S4) and the permutation-null mean (S10).

The methodological criterion of Version 2 is retained: a statistic carries sine-kernel content non-tautologically if and only if its value under the null differs from its value under the sine kernel (Remark 15.6). Version 3 adds the corresponding criterion for weights: every diagonal weight with a multiplicative symbol collapses onto $1/\zeta(1+s)$ (Thm. 14.1), so informative weights must break the Euler factorisation.

2. Standing results used as inputs

Established in separate work by the author, stated without proof, no novelty claimed here.

Remark 2.1. (S1) $c\sqrt{N} \le \lambda_{\max}(D_N) \le C_\varepsilon N^{1/2+\varepsilon}$ with Cauchy interlacing corollaries. *Context: Rem. 18.5.*

(S2) Trace-norm convergence of G_N/N^2 to a compact limit operator. *Context.*

(S3) Exact Chinese-remainder formula for the entries of Q_N. *Context; source of (S4).*

(S4) The tensor factorisation

$$\widehat{K}(r,q) = \prod_p L_p(v_p(r), v_p(q)) \tag{2.1}$$

over all of $\mathbb{N}\times\mathbb{N}$, explicit tridiagonal local factors, sign corrected. *Used actively as (G2).*

(S5) $C_{\mathrm{herm}}(\rho) = 2(\gamma^2-\beta^2)/(|\rho|^4|\zeta'(\rho)|^2) > 0$ for every nontrivial zero, without assuming RH. *Context; see Rem. 2.2.*

(S6) $z_0''(n) = -\frac{r^2}{2}\operatorname{Var}_{p|n}(\log p)$ for squarefree n with r prime factors, and $K(s) = \frac{\zeta(s)}{2}(S_0S_2 - S_1^2)$ with $S_2 = (\zeta'/\zeta)'$. *Context.*

(S7) Closed form of $z_\vartheta(n)$ as a finite cosine sum over ordered partitions via the cocycle $\Phi(\pi)$, with $A_r = 3(6r-11)/8$, $B_r = r(6r-11)/8$ exact for all r. *Context; source of the twisted action in Conj. 15.14.*

(S8) Blindness to Θ of the whole tower of *fixed* transversal jets, with complete proof. *Context; the restriction to fixed jets leaves the scaling window open.*

(S9) Unconditional equivalence between RH and tail stability of a two-channel positive logarithmic deformation of the ordinal-prime transform. *Context.*

(S10) $\mathbb{E}_\pi[E_N]/N = 1/\pi^2 + M(N)^2/(3N) + o(1)$ with $E_N = \frac{1}{N}\sum_{n\le N} M(n)^2$. *Used actively.*

Remark 2.2. At the extremal abscissa $s = 2\Theta$ all contributions come from zeros with $\beta = \Theta$ and carry the positive coefficient of (S5); a sum of positive terms does not cancel, so cancellation between distinct pairs cannot occur there. What remains open is the case of a non-attained supremum, requiring a limiting formulation.

3. Three spaces and the maps between them

Definition 3.1. Let $A_N = \{a_1 < \cdots < a_m \le N : q(a_j) = 1\}$, $m = |A_N|$. Set

$$H_N = \mathbb{R}^N \qquad \text{(full space; } D,\ U_h\text{)}, \tag{3.1}$$

$$H_{\mathrm{sf}} = \mathbb{R}^m \qquad \text{(squarefree space)}, \tag{3.2}$$

$$H_{\mathrm{exc}} = \mathbb{R}^Z \qquad \text{(excursion space)}. \tag{3.3}$$

Let $Je_j = e_{a_j}$, so $J^*J = I$ and $P = JJ^* = \operatorname{diag}(q)$, and let F be the canonical incremental matrix

$$F_{n,j} = \Delta Q_j(n)\,\mathbf{1}_{E_j}(n), \tag{3.4}$$

with $Q_j = |M|$ on E_j (Definition 5.3). Throughout, $s \in \{\pm 1\}^Z$ denotes excursion signs in this canonical basis and s_μ the signs of the Möbius path, so that $\mu = Fs_\mu + x_{\mathrm{term}}$.

Remark 3.2. The three spaces retain different information. H_N retains position, triangularity, divisibility and boundary data. H_{sf} retains signs and the integer labels a_j; if the labels are discarded, divisibility is lost. H_{exc} retains only block orientations: all intrablock variation, gcd classes, valuations and individual positions are destroyed. Since $P\mu = \mu$, the inclusion J loses nothing for the vector μ itself, but it does lose operator information about perturbations off the mask.

Proposition 3.3. *With $U_h^{\mathrm{sf}} = J^* U_h J$, $A_h = (U_h + U_h^*)/2$ and $x = Fs + x_{\mathrm{term}}$ on the fibre,*

$$C_N(h) = s^{\mathsf{T}}(F^* A_h F)s + 2s^{\mathsf{T}} F^* A_h x_{\mathrm{term}} + x_{\mathrm{term}}^{\mathsf{T}} A_h x_{\mathrm{term}}. \tag{3.5}$$

Proof. Substitute $x = Fs + x_{\mathrm{term}}$ into (1.2) and expand the symmetric form. □

Remark 3.4. Equation (3.5) is exact, but by Cor. 7.3 the excursion space is a severe compression and must be subordinate: it is used in Section 11 only for the reflection-odd component and as one possible family of stopping blocks. The primary space is H_N.

4. The three-constraint model

Definition 4.1. The orientation bits are $\varepsilon = (\varepsilon_n)_{n \in A_N} \in \{\pm 1\}^m$, $\varepsilon_n = \mu(n)$. With $m_\pm$ the number of $n \in A_N$ with $\mu(n) = \pm 1$ and $\bar{\varepsilon} = M(N)/m$, the *fixed-count sign bridge* π is uniform on the vectors of $\{\pm 1\}^m$ with m_+ entries $+1$.

Proposition 4.2. *Under π,*

$$\mathbb{E}_\pi[\varepsilon\varepsilon^{\mathsf{T}}] = (1-\beta)I + \beta J_m, \qquad \beta = \frac{m\bar{\varepsilon}^{\,2} - 1}{m-1}, \tag{4.1}$$

J_m the all-ones matrix. Consequently, for an orthogonal projection P' of rank d,

$$\mathbb{E}_\pi[\langle \varepsilon, P'\varepsilon\rangle] = (1-\beta)d + \beta\|P'\mathbf{1}\|^2 = d + O(1) \quad \text{if } M(N) = o(\sqrt{m}). \tag{4.2}$$

Proof. Diagonal entries are 1. For $n \neq n'$, counting ordered pairs of distinct positions gives $\big((m_+ - m_-)^2 - m\big)/(m(m-1))$, and $m_+ - m_- = m\bar{\varepsilon}$. Trace against P' using $\operatorname{tr} P' = d$, $\operatorname{tr}(P'J_m) = \|P'\mathbf{1}\|^2$; for the last claim $|\beta| \le 1/(m-1)$ once $M(N)^2 \le m$. □

Remark 4.3. The factorisation of $\mathrm{Cov}_\pi(M(n)^2, M(n')^2)$ through the three-block decomposition $(n, n'-n, N-n')$, with its $O(m)$ variance evaluation, is recorded separately. Only its structural consequence is used: the covariance kernel is separable of bounded rank. By Prop. 5.4 the third conditioning replaces this by an exact closed form.

5. The reflection fibre

Definition 5.1. Let $0 = z_0 < z_1 < \cdots < z_Z$ enumerate $\{n \le N : M(n) = 0\} \cup \{0\}$, and

$$E_j = \{n : z_{j-1} < n \le z_j\}, \qquad j = 1, \dots, Z, \tag{5.1}$$

with terminal block $E_{Z+1} = \{n : z_Z < n \le N\}$, empty iff $M(N) = 0$. Write $u_j = z_{j-1}$, $v_j = z_j$. The reflection group $\mathcal{R}_N \cong (\mathbb{Z}/2)^Z$ acts by $(\sigma\cdot\varepsilon)_n = \sigma_{j(n)}\varepsilon_n$, the terminal block held fixed. Holding it fixed is a conditioning on $M(N)$, which the three-constraint model imposes through the fixed sign count.

Proposition 5.2. *$M_{\sigma\cdot\varepsilon}(n) = \sigma_{j(n)} M_\varepsilon(n)$, so the quadratic profile $\Pi_N = (M(n)^2)_{n \le N}$ is $\mathcal{R}_N$-invariant; conversely two sign vectors on the same mask with the same profile and the same value of $M(N)$ differ by an element of $\mathcal{R}_N$. The fibre of Π_N conditioned on $M(N)$ is exactly the orbit. Without that conditioning, if $M(N) \neq 0$ the terminal block can also be reflected and the fibre of Π_N is an orbit of $(\mathbb{Z}/2)^{Z+1}$.*

Proof. For $n \in E_j$, $M_\varepsilon(z_{j-1}) = 0$ so $M_\varepsilon(n) = \sum_{z_{j-1}<k\le n} \varepsilon_k$ lies inside E_j and σ multiplies it by σ_j. Conversely equal profiles give the same zero set, hence the same blocks, and $|M|$ pointwise; inside a block M has constant sign, and the ratio of the two signs defines σ_j; increments determine the path. □

Definition 5.3. The excursion signs $s \in \{\pm1\}^Z$ record the sign of M in the interior of each E_j. The reflection null ν is uniform on $\{\pm1\}^Z$. Write $Q_j(n) = |M(n)|$ for $n \in E_j$, so that

$$M_s(n) = s_j Q_j(n), \qquad \mu_s(n) = s_j\, \Delta Q_j(n), \qquad n \in E_j. \tag{5.2}$$

Proposition 5.4. *Under ν, $\mathbb{E}_\nu[ss^{\mathsf T}] = I_Z$ exactly, hence for symmetric A*

$$\mathbb{E}_\nu[\langle s, As\rangle] = \operatorname{tr} A, \qquad \operatorname{Var}_\nu[\langle s, As\rangle] = 2\sum_{j\neq k} A_{jk}^2, \tag{5.3}$$

and for a projection $\widehat{P}$ of rank d, $\mathbb{E}_\nu = d$ and $\operatorname{Var}_\nu = 2(d - \sum_j \widehat{P}_{jj}^2)$.

Proof. The s_j are independent Rademacher. Write $\langle s, As\rangle = \operatorname{tr} A + \sum_{j\neq k} A_{jk} s_j s_k$ and use $\mathbb{E}_\nu[s_j s_k s_{j'} s_{k'}] = \mathbf{1}[\{j,k\} = \{j',k'\}]$; for a projection $\sum_{j,k} \widehat{P}_{jk}^2 = \operatorname{tr}\widehat{P} = d$. □

Remark 5.5. Conditioning on the mask removes the $1/\pi^2$ of (S10); conditioning on the count removes the drift $M(N)^2/(3N)$; conditioning in addition on the profile leaves exactly $\mathcal{R}_N$, whose null is isotropic without the $O(1)$ correction of (4.2). The third conditioning is where the null becomes parameter-free.

6. The divisor-defect Gram form on orientation bits

Definition 6.1. Let B be the divisor-defect matrix on A_N and $G = B^{\mathsf T}B$. G is *admissible* if (G1) it is real symmetric positive semidefinite and (G2) its kernel admits the tensor factorisation (2.1). The orientation-bit form is

$$\widehat{G} = F^{\mathsf T} G F, \quad \widehat{G}_{jk} = \sum_{a\in E_j}\sum_{b\in E_k} \Delta Q_j(a)\, G_{ab}\, \Delta Q_k(b), \qquad \widehat{G} = \sum_k \lambda_k \widehat{P}_k, \quad d_k = \operatorname{rank}\widehat{P}_k. \tag{6.1}$$

Proposition 6.2. *$\langle s, \widehat{G}s\rangle$ is invariant under $s \mapsto -s$. Under a single reflection $s_j \mapsto -s_j$ the terms incident to j, $2\widehat{G}_{jk}s_j s_k$ with $k \neq j$, change sign and all other terms are unchanged. Hence the form is not $\mathcal{R}_N$-invariant unless $\widehat{G}_{jk} = 0$ for all $k \neq j$, and it is not a function of the quadratic profile unless $\widehat{G}$ is diagonal.*

Proof. $s_j s_k$ changes sign under $s_j \mapsto -s_j$ for $k \neq j$, terms $s_k s_l$ with $k, l \neq j$ are unchanged, and $\widehat{G}_{jj}s_j^2$ is constant; by Prop. 5.2 functions of the profile are $\mathcal{R}_N$-invariant. □

7. Blindness, and its exact extension

Lemma 7.1. $\displaystyle\sum_{n\le N} M(n)^2 = \sum_{a,b\le N} \mu(a)\mu(b)\big(N + 1 - \max(a,b)\big).$

Proof. Expand $M(n)^2$ and exchange summation; the pair (a,b) occurs for $N+1-\max(a,b)$ values of n. □

Theorem 7.2. *Assume $M(N) = 0$. For all $j \neq k$,*

$$\sum_{a\in E_j}\sum_{b\in E_k} \mu(a)\mu(b)\big(N + 1 - \max(a,b)\big) = 0. \tag{7.1}$$

Hence $\sum_{n\le N} M(n)^2$ depends only on within-excursion two-point data.

Proof. For $j < k$ every $a \in E_j$ precedes every $b \in E_k$, so $\max(a,b) = b$ and the double sum factorises as $\big(\sum_{a\in E_j} \mu(a)\big)\big(\sum_{b\in E_k} \mu(b)(N+1-b)\big)$; the first factor is $M(z_j) - M(z_{j-1}) = 0$. The case $j > k$ is symmetric. □

Corollary 7.3. *No statistic factoring through Π_N detects cross-excursion correlation of μ.*

Proof. Such a statistic is $\mathcal{R}_N$-invariant by Prop. 5.2; by Thm. 7.2 the cross-excursion mass does not enter the energy, and by the action it is exactly the part on which $\mathcal{R}_N$ acts nontrivially. □

7.1. **Two channels and the half-constant extension.**

Lemma 7.4. *Let m_j be an interior cut of E_j, $E_j^L = (u_j, m_j]$, $E_j^R = (m_j, v_j]$, and $H_j = Q_j(m_j)$. Then*

$$\sum_{n\in E_j^L} \mu_s(n) = s_j H_j, \qquad \sum_{n\in E_j^R} \mu_s(n) = -s_j H_j, \qquad H_j \geq 1. \tag{7.2}$$

In particular the two halves are not independent channels: they have rank one.

Proof. Telescoping and $M(u_j) = M(v_j) = 0$ give the two sums; $H_j \geq 1$ because M does not vanish in the interior of an excursion. □

Theorem 7.5. *Let $j < k$ and let a kernel $K(a,b)$ be constant in a on each half of E_j, say $K(a,b) = K_L(b)$ for $a \in E_j^L$ and $K_R(b)$ for $a \in E_j^R$. Then*

$$\sum_{a\in E_j}\sum_{b\in E_k} \mu_s(a)\mu_s(b)K(a,b) = s_j H_j \sum_{b\in E_k} \mu_s(b)\big(K_L(b) - K_R(b)\big). \tag{7.3}$$

If $K_L = K_R$ the exact vanishing of Thm. 7.2 is recovered; otherwise the interaction factors through the single channel $s_j H_j$.

Proof. Split the a-sum over the two halves and apply Lemma 7.4 to each. □

Remark 7.6. Theorem 7.2 is the case $K_L = K_R$, because a kernel depending on $\max(a,b)$ loses all a-dependence when $j < k$. The shift kernel $K_h(a,b) = \mathbf{1}[b - a = h]$ is not constant on any half: it resolves $a = b - h$ at unit scale. Hence the shift lies outside the extended class, and this is the precise sense in which $C_N(h)$ escapes the excursion compression.

7.2. **The maximal blind class.**

Theorem 7.7. *Let E, F be blocks and P_E, P_F the projections onto vectors of zero sum. For a kernel K_{EF} the following are equivalent: $x^{\mathsf{T}} K_{EF} y = 0$ for all $x \perp \mathbf{1}_E$, $y \perp \mathbf{1}_F$; $P_E K_{EF} P_F = 0$; $K(a,b) = \alpha(a) + \beta(b)$ on $E \times F$. If the block of the shift contains r disjoint unit entries, its squared Frobenius distance to this class is*

$$r - \frac{r}{|E|} - \frac{r}{|F|} + \frac{r^2}{|E||F|}. \tag{7.4}$$

Proof. The first two conditions are the same statement. $P_E K P_F = 0$ iff K lies in the span of $\mathbf{1}_E \otimes \cdot$ and $\cdot \otimes \mathbf{1}_F$, which is the additively separable class. The distance is $\|P_E K P_F\|_F^2 = \|K\|_F^2 - \frac{\|\mathbf{1}^{\mathsf{T}} K\|^2}{|E|} - \frac{\|K\mathbf{1}\|^2}{|F|} + \frac{(\mathbf{1}^{\mathsf{T}} K \mathbf{1})^2}{|E||F|}$, and for r disjoint unit entries these are r, $r/|E|$, $r/|F|$ and $r^2/(|E||F|)$. □

Remark 7.8. Theorem 7.2 is the case $K(a,b) = \beta(b)$ for ordered blocks. The distance (7.4) is of order r, not a boundary error. Globally, U_h has H singular values equal to 1, so by Eckart–Young every kernel K_r of rank $r < H$ satisfies $\|U_h - K_r\|_{\mathrm{op}} \geq 1$ and $\|U_h - K_r\|_{\mathrm{HS}} \geq \sqrt{H - r}$: no separable approximation of sublinear rank is close in the norm needed for uniform observability.

8. The divisor–shift operator

8.1. What conjugation does not give.

Remark 8.1. D is lower triangular with $D^{-1}_{r,d} = \mu(r/d)\mathbf{1}[d \mid r]$, and

$$(DU_hD^{-1})_{r,m} = \sum_{\substack{a\mid r,\ a+h\le N\\ m\mid a+h}} \mu\Big(\frac{a+h}{m}\Big). \tag{8.1}$$

This is exact and gives no spectral gain: a similarity preserves the problem. The entries of the commutator,

$$[D,U_h]_{r,m} = \mathbf{1}[m>h]\mathbf{1}[m-h \mid r] - \mathbf{1}[r\le N-h]\mathbf{1}[m \mid r+h], \tag{8.2}$$

are not supported on primes dividing h: they compare the two distinct geometries $m-h \mid r$ and $m \mid r+h$.

8.2. The difference that localises.

Lemma 8.2 (Divisor–shift boundary lemma)**.** *Let $H = N-h$ and let $B^\circ_h = R_H(U_h - I)D$ be the restriction to the interior rows $1 \le n \le H$. Then*

$$(B^\circ_h)_{n,d} = \mathbf{1}[d \mid n+h] - \mathbf{1}[d \mid n], \qquad 1 \le n \le H, \tag{8.3}$$

while on the full operator the rows $r > H$ equal $-\mathbf{1}[d \mid r]$, so that there the columns $d \mid h$ do not vanish. For B°_h:

- *(i) If $d \mid h$ the column vanishes identically.*
- *(ii) If $d \nmid h$ the column is supported on the two disjoint classes $n \equiv -h$ and $n \equiv 0 \pmod d$, with signs $+1$ and -1.*
- *(iii) For every interval I of integers,*

$$\Big|\sum_{n\in I}\big(\mathbf{1}[d \mid n+h] - \mathbf{1}[d \mid n]\big)\Big| \le 1. \tag{8.4}$$

Proof. (8.3) is $D_{n+h,d} - D_{n,d}$. (i) If $d \mid h$ then $d \mid n+h \iff d \mid n$. (ii) Immediate. (iii) The sum equals the number of multiples of d in $I+h$ minus the number in I; two intervals of equal length contain numbers of multiples differing by at most one. □

Remark 8.3. This converts the multiplicative–additive interaction into residue-class boundaries of uniformly bounded interval discrepancy, with no regularity assumption on I. The bound 1 is optimal and improves the bound 2 obtained by estimating the two classes separately. Version 5 stated (8.3) for the full operator; that was incorrect on the boundary rows $r > H$ and is corrected here.

8.3. Structure and norms of the interior operator.

Theorem 8.4 (Maximal rank)**.** $\operatorname{rank} B^\circ_h = H = N-h$ *and* $\dim\ker B^\circ_h = h$*, as a map $\mathbb{R}^N \to \mathbb{R}^H$.*

Proof. Take the columns $d = h+r$, $1 \le r \le H$. For $n < r$ one has $0 < n+h < h+r$ and $n < h+r$, so neither $h+r \mid n+h$ nor $h+r \mid n$; for $n = r$, $h+r \mid r+h$ and $h+r \nmid r$. The $H \times H$ submatrix on these columns is therefore lower triangular with unit diagonal. □

Remark 8.5. The vanishing of the columns $d \mid h$ removes only $\tau(h)$ columns and produces no useful subspace: B°_h is surjective onto the interior coordinates.

Proposition 8.6 (Exact norms)**.** *For $d \nmid h$ let $a_d = \lfloor H/d \rfloor$ and $b_d = \lfloor N/d \rfloor - \lfloor h/d \rfloor$, the numbers of $n \le H$ with $d \mid n$ and with $d \mid n+h$. Then the column d has ℓ^1-norm and squared ℓ^2-norm $a_d + b_d$, the columns $d \mid h$ vanish, and*

$$\|B_h^\circ\|_{1\to 1} = \max_{d \nmid h} (a_d + b_d), \tag{8.5}$$

$$\|B_h^\circ\|_{\infty\to\infty} = \max_{n \le H} \big(\tau(n) + \tau(n+h) - 2\tau((n,h))\big), \tag{8.6}$$

$$\|B_h^\circ\|_{\mathrm{HS}}^2 = \sum_{n \le H} \big(\tau(n) + \tau(n+h)\big) - 2 \sum_{e \mid h} \Big\lfloor \frac{H}{e} \Big\rfloor. \tag{8.7}$$

Proof. For $d \nmid h$ the two classes are disjoint, giving the column norms. Row n is supported on divisors of exactly one of $n, n+h$; the common divisors are those of $(n, n+h) = (n,h)$. Finally $\sum_{n \le H} \tau((n,h)) = \sum_{e \mid h} \lfloor H/e \rfloor$. □

Proposition 8.7 (Residue-boundary form and variation bound)**.** *For $d \nmid h$ write $h = q_d d + r_d$ with $0 < r_d < d$ and*

$$\mathsf{A}_d(t) = \Big\lfloor \frac{t+h}{d} \Big\rfloor - \Big\lfloor \frac{h}{d} \Big\rfloor - \Big\lfloor \frac{t}{d} \Big\rfloor = \mathbf{1}[t \bmod d \ge d - r_d]. \tag{8.8}$$

Then $(B_h^\circ)_{t,d} = \mathsf{A}_d(t) - \mathsf{A}_d(t-1)$: every column is the oriented boundary of a periodic block of r_d consecutive residues. Consequently, for f on an interval $I = [u,v]$,

$$\Big| \sum_{n \in I} f(n) (B_h^\circ)_{n,d} \Big| \le \min(|f(u)|, |f(v)|) + \mathrm{Var}_I(f), \tag{8.9}$$

and the endpoint term disappears when I and $I+h$ contain equally many multiples of d.

Proof. $\lfloor (t+h)/d \rfloor = q_d + \lfloor (t+r_d)/d \rfloor$ gives (8.8), and the difference in t counts the new multiple of d in $(h, t+h]$ minus the new one in $(0,t]$. For (8.9) sum by parts from either end using the partial sums, which have modulus at most 1 by Lemma 8.2(iii). □

8.4. **CRT pullback of the shift.**

Lemma 8.8.

$$(D^{\mathsf{T}} U_h D)_{d,e} = \#\{n \le N - h : d \mid n,\ e \mid n+h\}, \tag{8.10}$$

which vanishes unless $(d,e) \mid h$. If $(d,e) \mid h$ there is a unique class $c \bmod [d,e]$ and

$$(D^{\mathsf{T}} U_h D)_{d,e} = \frac{N-h}{[d,e]} + \varepsilon_{d,e,h,N}, \qquad |\varepsilon_{d,e,h,N}| \le 1. \tag{8.11}$$

Proof. The entry counts n with $n \equiv 0 \pmod d$ and $n \equiv -h \pmod e$. Compatibility of the two congruences requires $0 \equiv -h \pmod{(d,e)}$; when it holds the solution set is a single class modulo $[d,e]$, whose count in an interval of length $N-h$ is the stated main term with error of modulus at most 1. □

Remark 8.9. The bulk is an explicit gcd/lcm kernel and all finite irregularity sits in a boundary term. Section 9 removes the bulk exactly.

9. Algebraic Haar on arbitrary trees, and exact bulk vanishing

Lemma 9.1. *Let $I = I_0 \sqcup I_1$ with masses $m_0, m_1 > 0$, $m = m_0 + m_1$, and*

$$\psi_I = \sqrt{\frac{m_1}{m_0 m}}\, \mathbf{1}_{I_0} - \sqrt{\frac{m_0}{m_1 m}}\, \mathbf{1}_{I_1}. \tag{9.1}$$

Then $\langle \psi_I, \mathbf{1}_I \rangle = 0$ and $\|\psi_I\|_2 = 1$. For any binary tree terminating in singletons, the normalised constant together with the ψ_I form an orthonormal basis.

Proof. $\langle \psi_I, \mathbf{1}_I \rangle = \sqrt{m_0 m_1/m} - \sqrt{m_0 m_1/m} = 0$ and $\|\psi_I\|_2^2 = m_1/m + m_0/m = 1$. Orthogonality of two Haar functions follows from disjointness of their supports or, when nested, from the mean-zero property on the child in which the finer one is supported. Counting gives a basis. □

Remark 9.2. No doubling measure, comparable children or geometric hypothesis is used. The identity is finite-dimensional algebra.

Theorem 9.3 (Exact vanishing of the CRT bulk)**.** *Let* $R_{d,e,h}(n) = \mathbf{1}[d \mid n]\mathbf{1}[e \mid n+h]$. *If* $(d,e) \nmid h$ *then* $R \equiv 0$. *Otherwise* $R = \mathbf{1}[n \equiv c \bmod q]$ *with* $q = [d,e]$, *and writing* $\#(I_j \cap (c \bmod q)) = m_j/q + \epsilon_j$ *with* $|\epsilon_j| < 1$,

$$\langle R_{d,e,h}, \psi_I \rangle = \sqrt{\frac{m_1}{m_0 m}}\,\epsilon_0 - \sqrt{\frac{m_0}{m_1 m}}\,\epsilon_1. \tag{9.2}$$

Moreover the boundary defects are additive over children:

$$\epsilon_I(c,q) = \epsilon_{I_0}(c,q) + \epsilon_{I_1}(c,q). \tag{9.3}$$

Proof. The compatibility statement is Lemma 8.8. Insert the counts into (9.1): the terms m_j/q contribute $\frac{1}{q}(\sqrt{m_0 m_1/m} - \sqrt{m_0 m_1/m}) = 0$ because ψ_I has mean zero, leaving (9.2). Additivity holds because the counts add and $m_I = m_{I_0} + m_{I_1}$. □

Remark 9.4. The irregularity of the cut is not an error term: it *is* the cocycle (ϵ_0, ϵ_1). By (9.3) the Haar coefficients are martingale differences of a single boundary cocycle, which opens a telescoping across scales that the excursion compression cannot see.

Proposition 9.5 (Tree-independent energy)**.** *For a residue class* R *and* $k_i = |R \cap I_i|$,

$$\langle \mathbf{1}_R, \psi_I \rangle = \frac{m_1 k_0 - m_0 k_1}{\sqrt{m_0 m_1 (m_0 + m_1)}}. \tag{9.4}$$

For any tree whose Haar functions complete the constant on $\{1, \dots, H\}$ *and any column* $d \nmid h$ *of* B_h°,

$$\sum_I \big|\langle (B_h^\circ)_{\bullet d}, \psi_I \rangle\big|^2 = a_d + b_d - \frac{(b_d - a_d)^2}{H}, \tag{9.5}$$

independently of the tree.

Proof. (9.4) is (9.1) evaluated on counts. Parseval gives $\sum_I |\langle f, \psi_I \rangle|^2 = \|f\|^2 - H^{-1}|\sum f|^2$; for the column, $\|f\|^2 = a_d + b_d$ and $\sum f = b_d - a_d$. □

Warning 9.6. A tree aligned with residue boundaries can concentrate the energy in few coefficients but cannot reduce it: irregularity produces sparsity, not coercivity. Any new estimate must use simultaneously the inclusion–exclusion signs, the compatibilities $(d,e) \mid h$ and the parent–children coherence (9.3).

Remark 9.7. The shift matrix in this basis is exact:

$$\langle \psi_I, U_h \psi_J \rangle = \sum_{\alpha,\beta \in \{0,1\}} c_{I,\alpha} c_{J,\beta}\, E_h(I_\alpha, J_\beta), \qquad E_h(X,Y) = \#\{n \in X : n + h \in Y\}, \tag{9.6}$$

with $c_{I,0}, c_{I,1}$ the two coefficients of (9.1). There is an entry only when an edge joins the supports; cancellation depends on the 2×2 incidence table, not on regular lengths.

10. Observability

Definition 10.1. Let $V = PH_N$, $L = D_\Lambda P|_V$ for a set Λ of divisor rows, and $A = PA_hP|_V$.

Theorem 10.2 (Exact observability criterion)**.** *There exists $\kappa < \infty$ with*

$$|x^{\mathsf T}Ax| \le \kappa \|Lx\|^2 \qquad \text{for all } x \in V \tag{10.1}$$

if and only if $\ker L \subseteq \ker A$. *In that case $A = L^{\mathsf T}BL$ with $B = (L^+)^{\mathsf T}AL^+$ and the best constant is*

$$\kappa_* = \big\|(L^{\mathsf T}L)^{\dagger/2} A (L^{\mathsf T}L)^{\dagger/2}\big\|_{\mathrm{op}}. \tag{10.2}$$

Proof. Sufficiency: write $x = x_0 + x_1$ with $x_0 \in \ker L$ and $x_1 \perp \ker L$. Then $Ax_0 = 0$ gives $x^{\mathsf T}Ax = x_1^{\mathsf T}Ax_1$, while $\|Lx\| = \|Lx_1\| \ge \sigma\|x_1\|$ with σ the least positive singular value, so (10.1) holds with $\kappa = \|A\|/\sigma^2$.

Necessity: let $z \in \ker L$ and $x = z + ty$. Then $\|Lx\|^2 = t^2\|Ly\|^2$ while $x^{\mathsf T}Ax = z^{\mathsf T}Az + 2t\, z^{\mathsf T}Ay + t^2 y^{\mathsf T}Ay$. Letting $t \to 0$ forces $z^{\mathsf T}Az = 0$, and then the linear term must be $O(t^2)$, forcing $z^{\mathsf T}Ay = 0$ for all y, i.e. $Az = 0$. Formula (10.2) is the induced norm on the range. □

Remark 10.3. Without the kernel condition the best Frobenius fit is $B_F = (L^+)^{\mathsf T}AL^+$ with residual $R_F = A - P_{(\ker L)^\perp} A P_{(\ker L)^\perp}$, and in operator norm the exact dual is

$$\inf_B \|A - L^{\mathsf T}BL\|_{\mathrm{op}} = \sup\{|\operatorname{tr}(AZ)| : Z = Z^{\mathsf T},\ \|Z\|_* \le 1,\ LZL^{\mathsf T} = 0\}, \tag{10.3}$$

the constraint $LZL^{\mathsf T} = 0$ being exactly annihilation of the subspace $\{L^{\mathsf T}BL\}$. For an error $\varepsilon\|x\|^2$ the optimal pair (κ, ε) solves the semidefinite programme $-\kappa L^{\mathsf T}L - \varepsilon I \preceq A \preceq \kappa L^{\mathsf T}L + \varepsilon I$.

Theorem 10.4 (No sublinear observability)**.** *Fix $h \neq 0$. If q and $q+h$ are squarefree with $q > N/2$ and neither row q nor row $q+h$ lies in Λ, then $x = e_q \pm e_{q+h} \in \ker L$ with $|x^{\mathsf T}A_hx| = 1$ and $\|x\|^2 = 2$; hence any inequality of the form $|x^{\mathsf T}A_hx| \le \kappa\|Lx\|^2 + \varepsilon\|x\|^2$ forces $\varepsilon \ge 1/2$. Consequently, to achieve $\varepsilon < 1/2$ the set Λ must be a vertex cover of the shift graph on $(N/2, N]$, and since that graph has maximum degree 2 and, by Mirsky's theorem* [9]*, a positive density of edges, $|\Lambda| = \Omega_h(N)$. No nontrivial observability is possible with $o(N)$ divisor rows.*

Proof. For $q > N/2$ the only multiple of q up to N is q itself, so column q of D is seen only by row q; the same for $q + h$. Hence $Lx = 0$ for the stated x, while $x^{\mathsf T}A_hx = \pm 2 \cdot \frac{1}{2} = \pm 1$ and $\|x\|^2 = 2$. The covering statement follows since each uncovered edge produces such an x; a graph of maximum degree 2 with $\Omega(N)$ edges needs $\Omega(N)$ vertices to cover. □

Corollary 10.5 (Quantitative form)**.** *With $\mathfrak{S}_h$ the Mirsky density of* (13.3)*, any Λ achieving $\varepsilon < 1/2$ in Theorem 10.4 satisfies*

$$|\Lambda| \ge \frac{\mathfrak{S}_h}{4} N + o_h(N). \tag{10.4}$$

Proof. The shift graph on $(N/2, N]$ has $\frac{\mathfrak{S}_h}{2}N + o_h(N)$ edges by [9] and maximum degree 2, so every vertex cover has at least half as many vertices as there are edges. □

Remark 10.6. Centres, quarters and returns can therefore only work after restricting to a low-dimensional multiscale subspace; global identification is excluded outright. The dual formula (10.3) may equivalently be written with the constraint $QZQ = 0$, $Q = L^+L$.

11. Centres, divisor rows and the Möbius line

11.1. **The central dichotomy.**

Lemma 11.1 (Central dichotomy)**.** *Let L_j be any linear functional of the values or increments of M on E_j, fixed before the orientation is known. Then*

$$L_j(M_s) = s_j L_j(Q_j) = s_j \kappa_j, \tag{11.1}$$

so either $\kappa_j = 0$ and L_j vanishes on the whole fibre, or $\kappa_j \neq 0$ and $s_j = \operatorname{sgn}(L_j(M_s)/\kappa_j)$ is read off directly. There is no third possibility.

Proof. Immediate from (5.2). □

Remark 11.2. With m_j^{sf} the support centre, i.e. the vertex after ℓ_j of the $2\ell_j$ squarefree steps of E_j, one has $1 \le Q_j(m_j^{\mathrm{sf}}) \le \ell_j$ and $Q_j(m_j^{\mathrm{sf}}) \equiv \ell_j \pmod 2$, so the central path value never vanishes and always reads the bit. The support centre is the canonical choice: it is squarefree, so a divisor row evaluated there necessarily contains its own diagonal term, and it avoids spurious plateaux with $\mu(n) = 0$.

Warning 11.3. Augmenting the Gram form with central path readings gives $C_{\mathrm{path}} = -2\,\mathrm{diag}(h_1, \dots, h_Z)$ and hence $G^+ = V^{\mathsf{T}} A V + C_{\mathrm{path}}^{\mathsf{T}} W C_{\mathrm{path}}$ with $\ker G^+ = \ker V \cap \ker C_{\mathrm{path}} = \{0\}$, since $h_j \neq 0$. The kernel disappears, but any J_h then admits $J_h = C_{\mathrm{path}}^{\mathsf{T}}(C_{\mathrm{path}}^{-\mathsf{T}} J_h C_{\mathrm{path}}^{-1}) C_{\mathrm{path}}$ with zero remainder. This is tautological: the whole difficulty has been moved inside a diagonal bit reader.

11.2. **Sampling the divisor equation at the centre.**

Definition 11.4. With $m_j = m_j^{\mathrm{sf}}$ and q_k the canonical positive increment of excursion k, set

$$C_{jk}^{\mathrm{div}} = \sum_{\substack{d \mid m_j \\ d \in E_k}} q_k(d). \tag{11.2}$$

Proposition 11.5. *$C_{jk}^{\mathrm{div}} = 0$ for $k > j$, so C^{div} is lower triangular. If moreover $u_j \ge m_j/2$, then $C_{jj}^{\mathrm{div}} = q_j(m_j) \in \{\pm 1\}$.*

Proof. For $k > j$ every element of E_k exceeds m_j, so no divisor of m_j lies in E_k. If $u_j \ge m_j/2$ then every proper divisor of m_j is $\le m_j/2 \le u_j$, hence outside $E_j = (u_j, v_j]$; only $d = m_j$ remains and m_j is squarefree. □

Proposition 11.6 (Logarithmic exceptional set)**.** *The hypothesis $u_j \ge m_j/2$ of Prop. 11.5 holds for all but at most $\lfloor \log_2 N \rfloor$ indices j.*

Proof. Since $m_j \le v_j$, the inequality $u_j \ge v_j/2$ implies $u_j \ge m_j/2$, so the hypothesis can fail only when $v_j > 2u_j$. Since $M(1) = 1$ and $M(2) = 0$, $z_1 = 2$; thus $u_1 = 0$ and $u_j \ge 2$ for $j \ge 2$. Let $2 \le j_1 < \cdots < j_r$ be the failing indices with $j \ge 2$. Using $u_{j+1} = v_j$ and monotonicity of u, $N \ge v_{j_r} > 2u_{j_r} > 2^2 u_{j_{r-1}} > \cdots > 2^r u_{j_1} \ge 2^{r+1}$, so $r + 1 \le \lfloor \log_2 N \rfloor$. Adding the possible failure at $j = 1$ gives at most $\lfloor \log_2 N \rfloor$ failing indices. □

Theorem 11.7. *Let $\mathcal{J}$ be the set of indices satisfying $u_j \ge m_j/2$. In orientation coordinates, treating the row $n = 1$ affinely by $w(1) = 1$,*

$$C^{\mathrm{div}} s_\mu = 0, \qquad \operatorname{rank} C^{\mathrm{div}} \;\ge\; |\mathcal{J}| \;\ge\; Z - \lfloor \log_2 N \rfloor. \tag{11.3}$$

In particular $\dim\ker C^{\mathrm{div}} \le \lfloor \log_2 N \rfloor + 1$ and s_μ lies in that kernel.

Proof. Row j of $C^{\mathrm{div}} s_\mu$ equals $\sum_{d \mid m_j} \mu(d)$, which vanishes for $m_j > 1$; the affine treatment of $d = 1$ fixes the remaining normalisation. The rank bound follows from lower triangularity together with Prop. 11.5 on $\mathcal{J}$ and Prop. 11.6. □

Corollary 11.8 (Compressed selection)**.** *The number of $s \in \{\pm 1\}^Z$ satisfying the central divisor equations, with the affine normalisation of the row $n = 1$, is at most $2^{\lfloor \log_2 N \rfloor + 1} \le 2N$.*

Proof. By Theorem 11.7 the solution set is an affine subspace of dimension $k \le \lfloor \log_2 N \rfloor + 1$. It projects injectively onto some k coordinates, so its points with entries ± 1 are determined by those k entries, giving at most 2^k of them. □

Corollary 11.9 (Completion to unique selection)**.** *Assume $M(N) = 0$. With the canonical F of (3.4), $C^{\mathrm{div}}_{jk} = (DF)_{m_j,k}$, so C^{div} consists of rows of DF, which has rank Z. Hence at most $\lfloor \log_2 N \rfloor + 1$ further rows of DF complete C^{div} to a system of rank Z, whose unique solution with the normalisation of the row $n = 1$ is s_μ.*

Proof. $(DF)_{r,k} = \sum_{d|r} F_{d,k}$, and for $r = m_j$ this is (11.2). D is unitriangular and F has Z nonzero columns with disjoint supports, so $\operatorname{rank} DF = Z$; extend a basis of the row space of C^{div} to one of DF, which needs at most $Z - \operatorname{rank} C^{\mathrm{div}} \le \lfloor \log_2 N \rfloor + 1$ rows. Since $DFs_\mu = D\mu = e_1$, every row $r \ne 1$ vanishes at s_μ and the row $r = 1$ fixes the normalisation. □

Remark 11.10. This is a near-minimal compression of Möbius inversion inside the fibre: $Z + O(\log N)$ divisor equations instead of N select the Möbius orientation among 2^Z. It works inside a fibre that already fixes $|M(n)|$ for every n, so it carries no information about the size of M.

Remark 11.11. This is a sparse orientational selector: roughly one divisor equation per excursion, chosen canonically at its centre, isolating the Möbius line up to a kernel of logarithmic dimension. For the full divisor operator these rows are already present, $C^{\mathrm{div}} = RV$ with R a row selector, so they do not reduce $\ker V$; for a truncation V_y they reduce the kernel exactly when $C^{\mathrm{div}}|_{\ker V_y} \ne 0$.

Warning 11.12. Isolating the Möbius line does not bound the shift on it. If the central rows reduce the kernel to $\operatorname{span}\{s_\mu\}$, then estimating J_h on that kernel is exactly estimating the cross-excursion Möbius correlation. Centres help as an arithmetic selector, not yet as an estimator.

11.3. **Conditioning the two-point expansion on central data.**

Proposition 11.13. *Write the expansion* (3.5) *in the orientation bits as*

$$C_N(h; s) = A_{N,h} + \ell_h^{\mathsf{T}} s + s^{\mathsf{T}} J_h^{\mathrm{off}} s, \tag{11.4}$$

and split $s = (a, u)$ with $a \in \{\pm 1\}^F$ fixed and u free. Then

$$C_N(h; s) = A' + (\ell_U + 2J_{UF}\, a)^{\mathsf{T}} u + u^{\mathsf{T}} J_{UU}\, u, \qquad A' = A_{N,h} + \ell_F^{\mathsf{T}} a + a^{\mathsf{T}} J_{FF}\, a. \tag{11.5}$$

Proof. Write J_h^{off} in blocks indexed by F and U and use its symmetry, so that the two mixed blocks contribute $2a^{\mathsf{T}} J_{FU} u$. □

Remark 11.14. Conditioning on all signed central values fixes s, by Lemma 11.1, and the fibre is a singleton: every Walsh coefficient disappears only because the full vector has been supplied. Conditioning on central magnitudes $|M(m_j)|$ changes nothing, since they are profile data. Conditioning on a proper subset F converts fixed–free quadratic terms into linear ones but leaves the quadratic form J_{UU} on the unobserved bits intact. No quantitative reduction follows without an independent bound on J_{UU}.

Remark 11.15. On orientations, a single nonvanishing odd evaluation already reads s_j, so quarter and octave evaluations are redundant by Lemma 11.1. On the shift, a coarse Haar truncation does not approximate A_h uniformly on vectors with entries in $\{-1, 0, 1\}$, which may oscillate at unit scale; deterministic control of the remainder requires cells of size $O(h)$, that is, essentially full resolution for fixed h. Proposition 17.4 makes this exact.

12. Intrablock identities

Lemma 12.1. *Let $x_1, \dots, x_L$ have zero sum, $P_t = \sum_{r \le t} x_r$ extended by zero outside $[0, L]$, and $A_h = \sum_t P_t P_{t+h}$. Then*

$$C_h := \sum_t x_t x_{t+h} = 2A_h - A_{h-1} - A_{h+1}. \tag{12.1}$$

Moreover

$$C_0 + 2\sum_{h \ge 1} C_h = 0, \qquad \sum_{t=1}^{L-1} P_t^2 = -\sum_{h \ge 1} h C_h. \tag{12.2}$$

Proof. Expand $x_t = P_t - P_{t-1}$ and collect the four autocorrelations. The first identity in (12.2) is $\sum_{h \in \mathbb{Z}} C_h = (\sum_t x_t)^2 = 0$. For the second, apply (12.1) for all $h \in \mathbb{Z}$ and reindex: $\sum_h |h| C_h = \sum_h A_h(2|h| - |h+1| - |h-1|) = -2A_0$, since the bracket vanishes for $h \ne 0$ and equals -2 at $h = 0$; halve. □

Remark 12.2. Inside an excursion $P_t P_{t+h} = |P_t||P_{t+h}|$, so the intrablock autocorrelation is already determined by the absolute profile. The centre adds nothing there.

Remark 12.3 (The centre does not determine a lag). The two positive excursions of length 8

$$(+,+,+,-,+,-,-,-) \quad \text{and} \quad (+,+,-,+,-,+,-,-) \tag{12.3}$$

have paths $(1,2,3,2,3,2,1,0)$ and $(1,2,1,2,1,2,1,0)$, hence the same central value 2, but $C_1 = 1$ and $C_1 = -3$ respectively. Length, zero sum and central height do not control even the first lag.

Lemma 12.4 (Left–right splitting)**.** *With m the centre and $C_h = C_h^{LL} + C_h^{RR} + C_h^{\times}$, only indices $m - h < t \le m$ can cross, so $|C_h^{\times}| \le h$ and, summing over excursions, $|\sum_j C_{h,j}^{\times}| \le hZ$.*

Proof. A pair $(t, t+h)$ crosses m iff $t \le m < t + h$. □

Remark 12.5. Lemma 12.4 is $o(N)$ precisely when $Z = o(N)$, which is not proved anywhere in this project; see (Q5). The other two parts can be of the size of the excursion length: the centre isolates a layer of width h only.

Lemma 12.6 (Discrete coarea)**.**

$$\mu(n) = \sum_{\ell \in \mathbb{Z}} \Big(\mathbf{1}[M(n-1) \le \ell < M(n)] - \mathbf{1}[M(n) \le \ell < M(n-1)] \Big). \tag{12.4}$$

Consequently $C_N(h)$ is a sum of oriented intersections of level-set boundaries separated by h.

Proof. If $\mu(n) = +1$ the first indicator contributes once and the second is empty; if $\mu(n) = -1$ the roles are exchanged; if $\mu(n) = 0$ both are empty. □

Proposition 12.7 (Global second difference)**.** *With $H = N - h$ and $A_k(L) = \sum_{n=1}^{L} M(n)M(n+k)$,*

$$C_N(h) = 2A_h(H-1) - A_{h-1}(H-1) - A_{h+1}(H-1) + M(H)\mu(N). \tag{12.5}$$

For a partition into B blocks of zero sum, i.e. between returns of M to a common level, not necessarily zero, with local prefixes P_j extended by zero and $A_j(k) = \sum_n P_j(n)P_j(n+k)$,

$$C_N(h) = \sum_j \big(2A_j(h) - A_j(h-1) - A_j(h+1)\big) + R_\partial + R_{\text{tail}}, \qquad |R_\partial| \le h(B-1), \tag{12.6}$$

with $|R_{\text{tail}}|$ bounded by the length of the terminal tail.

Proof. Expand $\mu(n) = M(n) - M(n-1)$ in $\sum_{n \le H} \mu(n)\mu(n+h)$: the four sums are $A_h(H)$, $A_{h-1}(H)$, $A_{h+1}(H-1)$ and $A_h(H-1)$, and $A_h(H) - A_{h-1}(H) = A_h(H-1) - A_{h-1}(H-1) + M(H)\mu(N)$. The block version is Lemma 12.1 on each block, with at most h crossing pairs per cut. □

Remark 12.8. The uncontrolled term is the curvature at scale h of the local path autocorrelations. Centres and hitting times can shorten the tail, but each new cut costs up to h pairs, and no universal choice makes both the number of cuts and the curvatures small. In the coarea form $C_N(h) = \sum_{\ell,m} \sum_{n\le H} \sigma_\ell(n)\sigma_m(n+h)$, up- and down-crossings balance globally in closed blocks, but their shifted products do not pair: the shift selects particular times, not totals.

13. Exact local structure at each prime

Proposition 13.1. *Fix a prime p and $t = v_p(h)$. Counting residues modulo p^2, the admissible states $(v_p(n), v_p(n+h)) \in \{0,1\}^2$ and their multiplicities are*

t	*states and multiplicities*	*local sign*
$t=0$	$(0,0)$*:* p^2-2p*;* $(1,0)$*,* $(0,1)$*:* $p-1$ *each*	$+,-,-$
$t=1$	$(0,0)$*:* p^2-p*;* $(1,1)$*:* $p-2$	$+$
$t\ge 2$	$(0,0)$*:* p^2-p*;* $(1,1)$*:* $p-1$	$+$

The local squarefree density of the pair is $1-2/p^2$ if $p^2 \nmid h$ and $1-1/p^2$ if $p^2 \mid h$, which is the Mirsky structure [9]*. For any finite set of primes P,*

$$\prod_{p\in P} (-1)^{v_p(n)+v_p(n+h)} \mathbf{1}[v_p(n), v_p(n+h) \le 1] \tag{13.1}$$

is an exact identity, not an independence hypothesis.

Proof. For $p \nmid h$ the classes $n \equiv 0$ and $n \equiv -h \pmod p$ are distinct, giving $(p-2)p$ residues with $(0,0)$ and $p-1$ each with $(1,0)$, $(0,1)$; the two excluded residues are $p^2 \mid n$ and $p^2 \mid n+h$. For $p\|h$, $p \mid n$ forces $p \mid n+h$, and among the p residues with $p \mid n$ one must delete $n \equiv 0$ and $n \equiv -h$, distinct modulo p^2, leaving $p-2$. For $p^2 \mid h$ the two deleted residues coincide, leaving $p-1$. Signs are $(-1)^{v_p(n)+v_p(n+h)}$; the densities follow by counting the excluded residues. CRT makes the joint count over distinct primes multiplicative. □

Remark 13.2. At the level of residues the shift does not break the local structure: every prime participates, with a universal off-diagonal factor for $p \nmid h$ and a diagonal factor for $p \mid h$, and CRT is exact. What fails on globalising is the cut $n \le N$, the infinite product and the parity of large primes appearing to the first power. Proposition 13.5 shows that at the level of valuations the shift does break it.

Proposition 13.3 (Signed local sums and Mirsky density)**.** *With $w_p(n) = (-1)^{\mathbf{1}[p|n]+\mathbf{1}[p|n+h]}\mathbf{1}[p^2 \nmid n]\mathbf{1}[p^2 \nmid n+h]$, the sum of $w_p(n)$ over $n \bmod p^2$ equals p^2-4p+2, p^2-2 or p^2-1 according as $v_p(h) = 0$, 1 or ≥ 2. With $q(n) = \sum_{r^2\mid n} \mu(r)$,*

$$\sum_{n\le H} q(n)q(n+h) = H \sum_{\substack{r,s \\ (r,s)^2\mid h}} \frac{\mu(r)\mu(s)}{[r^2,s^2]} + \sum_{\substack{r,s \\ (r,s)^2\mid h}} \mu(r)\mu(s)\,\epsilon_{r,s}, \qquad |\epsilon_{r,s}| < 1, \tag{13.2}$$

the sums over $r \le \sqrt{H}$, $s \le \sqrt{H+h}$, and the main term has density

$$\mathfrak{S}_h = \prod_{p^2\mid h} \Big(1-\frac{1}{p^2}\Big) \prod_{p^2\nmid h} \Big(1-\frac{2}{p^2}\Big) > 0. \tag{13.3}$$

Proof. The signed sums follow from the table of Proposition 13.1. For (13.2), expand both q factors; the system $r^2 \mid n$, $s^2 \mid n+h$ is solvable iff $(r,s)^2 \mid h$, and then counts one class modulo $[r^2,s^2]$. The density is the product of the local densities of Proposition 13.1. □

Remark 13.4. Mirsky [9] controls the error in (13.2) by a power saving, recalled as $O_h(H^{2/3+o(1)})$ and not verified here. The tail $\sum_{n\le N} |q(n)-q_P(n)| \le \sum_{P<p\le\sqrt{N}} \lfloor N/p^2 \rfloor$ is controlled, but it controls only large squares, not the parity of large primes appearing to the first power.

Proposition 13.5 (The shift does not descend to valuations)**.** *Let $p \nmid h$ and choose units $u, v \bmod p$ with $u \equiv -h$ and $v \not\equiv -h \pmod p$. Then $v_p(u) = v_p(v) = 0$, while $v_p(u+h) \geq 1$ and $v_p(v+h) = 0$. Hence translation by h is not well defined on the quotient remembering only $v_p(n)$, and there is no exact intertwiner between valuation-indexed local factors such as the L_p of (S4) and the shift without adjoining residue information $n/p^{v_p(n)} \bmod p^k$.*

Proof. Immediate. □

14. Multiplicative weights collapse

Theorem 14.1. *Let $m_d(x) = \sum_{n\leq x,\ (n,d)=1} \mu(n)/n$. For $p \nmid d$,*

$$m_d(x) = m_{dp}(x) - \frac{1}{p}\, m_{dp}(x/p), \qquad m_{dp}(x) = \sum_{k\geq 0} p^{-k} m_d(x/p^k), \tag{14.1}$$

the second sum being finite. Consequently

$$m_d(x) = \sum_{\substack{e\geq 1 \\ \mathrm{rad}(e)\mid d}} \frac{1}{e}\, m_1(x/e), \tag{14.2}$$

and with $G_{ab} = 2N/[a,b]$,

$$\frac{\langle \mu, G\mu\rangle}{2N} = \sum_{d \text{ squarefree}} \varphi(d) \left(\sum_{\mathrm{rad}(u)=d} \frac{m_1(N/u)}{u} \right)^2. \tag{14.3}$$

In Mellin variables the local operator has symbol $1-p^{-1-s}$ and the full product has symbol $1/\zeta(1+s)$.

Proof. Split $\{(n,d) = 1\}$ according to $p \mid n$; for $n = pn'$ squarefree with $(n', dp) = 1$ one has $\mu(n) = -\mu(n')$, giving the first identity. The Neumann series inverts it and terminates because $m_d(x/p^k) = 0$ for $p^k > x$. Composing over $p \mid d$ gives (14.2). For (14.3), write $\gcd(a,b) = \sum_{d\mid a,\ d\mid b} \varphi(d)$ to obtain $\langle \mu, G\mu\rangle/2N = \sum_d \varphi(d) d^{-2} \mu(d)^2 m_d(N/d)^2$; substituting (14.2) with $u = de$, the conditions $\mathrm{rad}(e) \mid d$, $d \mid u$ and d squarefree force $d = \mathrm{rad}(u)$, so each u lies in exactly one block and $1/e = d/u$. The Mellin symbol follows from $\sigma_p f(x) = f(x/p) \leftrightarrow p^{-s}$. □

Corollary 14.2. *Expanding G along its diagonals and using $\gcd(a, a+h) = \gcd(a,h)$,*

$$2\sum_{h\geq 1} \sum_{a} \mu(a)\mu(a+h) \frac{\gcd(a,h)}{a(a+h)} = \sum_d \varphi(d) \Big(\sum_{\mathrm{rad}(u)=d} \frac{m_1(N/u)}{u} \Big)^2 - \sum_{a\leq N} \frac{\mu(a)^2}{a^2}, \tag{14.4}$$

so the gcd-weighted Chowla aggregate is determined by m_1, i.e. by $1/\zeta(1+s)$. The left side is in particular bounded below by $-\zeta(2)/\zeta(4)$ since the right side is a sum of squares.

Proof. Immediate from Theorem 14.1 and $G_{a,a+h} = 2N \gcd(a,h)/(a(a+h))$. □

Warning 14.3 (Scope of the collapse)*.* Theorem 14.1 proves collapse for the gcd/lcm kernel. The mechanism, closure of the divisor lattice, suggests that diagonal weights with a multiplicative symbol collapse in the same way; this is heuristic and is not proved here in general. The claim made in Version 4, that every weight without a multiplicative symbol is strictly harder than $C_N(h)$ with no intermediate case, is incorrect in that generality and is withdrawn. The proof of collapse uses $\mu(da') = \mu(d)\mu(a')$ and uniqueness of the radical, i.e. commutativity of the Euler product; a non-commutative deformation such as the cocycle $\Phi(\pi)$ of (S7) is the only place where the argument does not apply. That escape is a fixed jet in ϑ, and (S8) closes fixed jets; only the scaling window $\vartheta \asymp \vartheta_c(N)$ survives.

15. Scale-resolved statistic and sine-kernel content

Definition 15.1. $\mathcal{E}_k = \langle s, \widehat{P}_k s\rangle$, $\mathcal{R}_k = \mathcal{E}_k/d_k$.

Proposition 15.2. *Under* ν, $\mathbb{E}_\nu[\mathcal{R}_k] = 1$ *exactly and* $\mathrm{Var}_\nu[\mathcal{R}_k] = 2d_k^{-2}(d_k - \sum_j \widehat{P}^2_{k,jj})$, *with* $\sum_k \mathcal{E}_k = Z = \sum_k d_k$.

Proof. Prop. 5.4 with $A = \widehat{P}_k$, and $\sum_k \widehat{P}_k = I_Z$. □

Lemma 15.3. *The orthogonal projection onto a contiguous block of* d *characters of a cyclic group of order* Z *has Dirichlet kernel, whose scaling limit with* d/Z *fixed is* $\sin(\pi x)/(\pi x)$ [8]. *Equivalently*

$$1 - R_2(x) = \Big(\frac{\sin \pi x}{\pi x}\Big)^2 = \int_{-1}^{1} (1 - |\tau|) e^{2\pi i \tau x}\, d\tau. \tag{15.1}$$

Warning 15.4. By Prop. 5.4 the reflection null has covariance I, which restricted to a band of characters is a sine kernel. The sine kernel is therefore the *null* of $\mathcal{R}_k$, not a consequence of it, and deriving it from flatness carries no evidential weight about the zeros of ζ.

Remark 15.5. Hypothesis (G2), i.e. (S4), makes the eigenbasis of G a character basis of the multiplicative structure, so the transported band gives a sine kernel in $\log n$, the variable in which $n^{i\gamma} = e^{i\gamma \log n}$ oscillates. This does not affect Warning 15.4.

Remark 15.6. A statistic carries sine-kernel content non-tautologically if and only if its null value is a number different from its sine-kernel value; the measured position between the two is the coefficient. The three quantities of Version 2 satisfying this are retained below.

Definition 15.7. For the superposition of p independent identically distributed spectra each of density $1/p$,

$$R_2^{(p)}(x) = 1 - \frac{1}{p}\Big(\frac{\sin(\pi x/p)}{\pi x/p}\Big)^2, \qquad Y_{jk}^{(p)} = \frac{1}{p}\Big(\frac{\sin(\pi(j-k)/p)}{\pi(j-k)/p}\Big)^2. \tag{15.2}$$

Proposition 15.8. *If* $\tilde r$ *is strictly monotone in* p *with* $\tilde r(\infty) = 2\ln 2 - 1$ *and* $\tilde r(1)$ *the unitary value, then a measured* $\tilde r > 2\ln 2 - 1$ *forces* $p < \infty$ *and hence a strictly positive sine-kernel fraction* $1/p$.

Proof. Strict monotonicity and the endpoint values. □

Lemma 15.9. $m_n = \displaystyle\int_0^1 (1-\tau)\tau^n\, d\tau = \frac{1}{(n+1)(n+2)}$, *and* $(1-\tau)_+$, *having compact support, is determined by* $\{m_n\}_{n\ge 0}$.

Proof. Elementary integration and determinacy of the Hausdorff moment problem. □

Lemma 15.10. *Under the Wigner surmises,* $\mathrm{Var}_{\mathrm{Poisson}}(s) = 1$, $\mathrm{Var}_{\mathrm{GOE}}(s) = 4/\pi - 1$, $\mathrm{Var}_{\mathrm{GUE}}(s) = 3\pi/8 - 1$.

Proof. For $p(s) = \frac{\pi}{2} s e^{-\pi s^2/4}$ and $p(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$, both normalised to $\langle s\rangle = 1$, ratios of Gaussian moments give $\langle s^2\rangle = 4/\pi$ and $3\pi/8$; the Poisson case is the exponential law. □

Remark 15.11. Recorded in separate work, not computed here, and used only to indicate the scale of the effects: a spacing ratio $\tilde r \approx 0.42$ for the spectrum of the smooth arithmetic term; relative freeness defects of about 12% at order two and 32% at order three; an energy deficit ratio of about 0.137 for the area energy and about 0.052 for the truncated logarithmic energy; a persistent bias of about 2.5%; and the systematic observation that $d = 1$ is the weakest anomaly. The reference values $2\ln 2 - 1$, ≈ 0.5307 and ≈ 0.5996 are recalled from [1] and not verified here.

Remark 15.12. The triangle moments give $m_3/m_2 = 0.6$, decreasing; Chebyshev-basis variance of linear statistics grows linearly [5], giving 1.5; the recorded defects give $32/12 \approx 2.67$. The observed growth is faster than linear where the triangle requires decay. The ratio is invariant under a global rescaling of the normalisation of relative defect, so the discrimination survives that convention.

Remark 15.13. $\mathcal{R}_N$ and $\widehat{G}$ are real, hence $\beta = 1$, whose form factor $2\tau - \tau\log(1+2\tau)$ on $\tau \le 1$ [8] is not a band indicator; the conjectural class for the zeros is $\beta = 2$ [10, 11].

Remark 15.14 (Withdrawn conjecture). Versions 1–4 conjectured that letting $U(1)^Z$ act by $s_j \mapsto e^{i\vartheta_j} s_j$ and transporting $\widehat{G}$ produces an orthogonal-to-unitary crossover at $\vartheta_c \asymp 1/Z$. As stated this is incorrect: by Proposition 16.4 a transported form $U_\vartheta^* \widehat{G} U_\vartheta$ is isospectral for every ϑ, and the scale $1/Z$ does not follow from [13]. See Question 16.5.

16. Audit results of Version 5

16.1. The energy kernel sits at $\alpha = 0$.

Proposition 16.1. *Let $K(\Delta) = 1/(2 + i\Delta)$ with $\Delta = \gamma - \gamma'$. Then*

$$K(\Delta) + K(-\Delta) = \frac{4}{4+\Delta^2} = \int_{-\infty}^{\infty} e^{-2|u|} e^{-iu\Delta}\, du. \tag{16.1}$$

Under $u = \alpha \log T$ the induced normalised density on α is

$$p_T(\alpha) = \log T\; T^{-2|\alpha|}, \qquad \mathbb{E}|\alpha| = \frac{1}{2\log T}, \qquad \Pr(|\alpha| > A) = T^{-2A}. \tag{16.2}$$

Proof. Sum the two fractions; $\int e^{-a|u|} e^{-iu\Delta} du = 2a/(a^2 + \Delta^2)$ with $a = 2$. Change variables; $\int p_T = 1$, and the moments are elementary exponential integrals. □

Remark 16.2. The hermitised kernel is Montgomery's smoothing weight [10] without the factor $T^{i\alpha(\gamma-\gamma')}$, so the scalar energy tests the pair correlation at $\alpha = 0$, smeared over width $1/\log T$. Evaluating the closed forms at the height of the 400th zero, $T \simeq 679.74$, gives $\mathbb{E}|\alpha| \approx 0.0767$ and $\Pr(|\alpha| > 1) \approx 2.16 \times 10^{-6}$. To reach $\alpha_0 > 1$ one needs the modulated family

$$K_{\alpha_0, T}(\Delta) = \frac{4}{4+\Delta^2}\, e^{i\alpha_0 \log T\, \Delta}. \tag{16.3}$$

The dependence of the energy on $\log N$ (Remark 18.1) supplies exactly this phase at $\log N = \alpha_0 \log T$; what it does not supply is localisation in height (Remark 18.2). The modulated family is available, the localised one is not.

Warning 16.3. The Montgomery variable α of this subsection is unrelated to the parameter α of the ordinal-prime programme behind (S9), for which separate work records $\sigma_c(\alpha) = 1 + \Theta$ and RH $\iff \sigma_c(\alpha) = 3/2$. The two must not be conflated.

16.2. Gauge obstruction to the crossover.

Proposition 16.4. *(i) A Hermitian matrix whose off-diagonal support is the edge set of a forest is conjugate, by a diagonal unitary, to a real symmetric matrix with nonnegative off-diagonal entries. In particular a Hermitian Jacobi matrix with entries $a_j e^{\pm i\phi_j}$ has the spectrum of the real Jacobi matrix with entries a_j.*

(ii) If $\widehat{G}^{(\vartheta)} = U_\vartheta^* \widehat{G} U_\vartheta$ with U_ϑ unitary, then $\operatorname{Spec} \widehat{G}^{(\vartheta)} = \operatorname{Spec} \widehat{G}$ for every ϑ.

Proof. (i) Root each tree and choose the phase of each vertex recursively along edges so that every edge entry becomes real and nonnegative; no cycle imposes a compatibility condition. (ii) Unitary similarity. □

Question 16.5. Construct a Hermitian deformation of $\widehat{G}$, or of a divisibility operator, on a graph with cycles, with phases whose fluxes around cycles are not coboundaries, for instance built from the cocycle $\Phi(\pi)$ of (S7). Only such a deformation can change the spectrum.

Heuristic 16.6. For a non-isospectral deformation H_ϑ with $A = \frac{d}{d\vartheta} H_\vartheta\big|_{\vartheta=0}$, the crossover scale is expected at

$$\vartheta_c \asymp \frac{\Delta_Z}{\sqrt{\mathbb{E}|\langle u_j, A u_k\rangle|^2}}, \tag{16.4}$$

Δ_Z the mean level spacing and u_j the eigenvectors, in the spirit of [13]. Depending on normalisation this may be Z^{-1}, $Z^{-1/2}$ or another scale; the coefficients A_r, B_r of (S7) alone do not decide it.

16.3. **Hermitian collision versus the Möbius energy constant.**

Proposition 16.7. *Assume RH and simple zeros, and let* $\beta_M = \sum_\rho |\rho\zeta'(\rho)|^{-2} = 2\sum_{\gamma>0} |\rho|^{-2}|\zeta'(\rho)|^{-2}$. *For every* $\gamma > 0$,

$$\frac{2}{|\rho|^2|\zeta'(\rho)|^2} - C_{\mathrm{herm}}(\rho) = \frac{1}{|\rho|^4|\zeta'(\rho)|^2}, \tag{16.5}$$

hence, whenever the sums converge,

$$\sum_{\gamma>0} C_{\mathrm{herm}}(\rho) = \beta_M - \sum_{\gamma>0} \frac{1}{|\rho|^4|\zeta'(\rho)|^2}. \tag{16.6}$$

Proof. With $\beta = \frac12$, $|\rho|^2 = \gamma^2 + \frac14$ and $\gamma^2 - \beta^2 = \gamma^2 - \frac14$, so $\frac{2}{|\zeta'|^2}\big(\frac{1}{|\rho|^2} - \frac{\gamma^2-1/4}{|\rho|^4}\big) = \frac{2}{|\zeta'|^2}\cdot\frac{1/2}{|\rho|^4}$. □

Remark 16.8. The pointwise ratio is $(\gamma^2 - \frac14)/(\gamma^2 + \frac14)$, approximately 0.9975005 at the first zero: the two constants are close but not identical. The identification of a recorded logarithmic estimator $\hat S_2$ with β_M requires RH, simplicity, convergence and a limit, and is not asserted here.

16.4. **The form $\mathcal{A}_0'$ is a Lévy area.**

Definition 16.9. With $M_\Omega(x) = \sum_{n\le x} \mu(n)\Omega(n)$ and $L(x) = \sum_{n\le x} \mu(n)\log n$, the form recorded in separate work is

$$\mathcal{A}_0'(N) = \frac12 \sum_{m<n\le N} \mu(m)\mu(n)\big(\Omega(m)\log n - \log m\,\Omega(n)\big). \tag{16.7}$$

Proposition 16.10.

$$\Delta\mathcal{A}_0'(N) = \frac{\mu(N)}{2}\big[M_\Omega(N-1)\log N - \Omega(N)L(N-1)\big], \tag{16.8}$$

and consequently

$$\mathcal{A}_0'(N) = \mathfrak{A}_N(M_\Omega, L), \qquad \mathfrak{A}_N(X,Y) = \frac12 \sum_{n\le N} \big(X(n-1)\,\Delta Y(n) - Y(n-1)\,\Delta X(n)\big), \tag{16.9}$$

the discrete signed area swept by the planar path $n \mapsto (M_\Omega(n), L(n))$.

Proof. The terms of (16.7) with $n = N$ give (16.8). Summing, with $\Delta M_\Omega(n) = \mu(n)\Omega(n)$ and $\Delta L(n) = \mu(n)\log n$, gives (16.9). □

Remark 16.11. A Lévy area is the antisymmetric second level of an iterated sum, so it is neither a product of two summatory functions, which is the symmetric part, nor a fixed-shift correlation. This is why the dichotomy proposed in session discussion (either a product controlled by RH, or a Chowla-type correlation) is false. Separate work records $\Delta\mathcal{A}_0'(N) = \mu(N)N/(2\log N) + o(N/\log N)$.

Proposition 16.12. *For* $\Re s > 2$ *let* $F(s) = \sum_{n\ge2} \mu(n) n^{1-s}/\log n$. *Then*

$$F'(s) = 1 - \frac{1}{\zeta(s-1)}, \qquad F(s) = \int_s^\infty \Big(\frac{1}{\zeta(u-1)} - 1\Big)du, \tag{16.10}$$

so F has logarithmic singularities at $s = 1 + \rho$.

Proof. Differentiate termwise: $F'(s) = -\sum_{n\geq 2}\mu(n)n^{1-s} = -(1/\zeta(s-1)-1)$. Since $F(s)\to 0$ as $s\to+\infty$, integrate from s to ∞. Simple poles of $1/\zeta(s-1)$ at $s-1=\rho$ integrate to logarithms. □

Remark 16.13. The precise obstacle on the Perron–Mellin route is then to show that the exact remainder of the Dirichlet series of $\mathcal{A}_0'$ continues far enough and does not cancel these singularities.

16.5. **Energies as aggregates of shifted correlations.**

Lemma 16.14. *With* $W_N(r) = \sum_{n=r}^{N} n^{-2}$,

$$\sum_{n\leq N}\frac{M(n)^2}{n^2} = \sum_{a\leq N}\mu(a)^2 W_N(a) + 2\sum_{h=1}^{N-1}\sum_{a\leq N-h}\mu(a)\mu(a+h)\,W_N(a+h). \tag{16.11}$$

Proof. Expand $M(n)^2$; the pair (a,b) contributes $\sum_{n=\max(a,b)}^{N} n^{-2}$. □

Remark 16.15. Lemma 7.1 and Lemma 16.14 show that both energies are weighted aggregates of shifted correlations. The logarithmically averaged two-point theorem [16] controls fixed h; these aggregates need uniformity up to $h \asymp N$, and that gap is not closed by naming the logarithmic theorem. The two recorded ratios belong to different kernels, about 0.137 for the area energy and about 0.052 for the truncated logarithmic energy, and are not comparable.

Remark 16.16 (Corrections ledger). Corrected in Version 5, in accordance with the archival principle: the pushed form (6.1), previously written without the incremental factors; the orientation basis (3.4), previously mixing reflections relative to μ with canonical signs; the conditioning on $M(N)$ in Proposition 5.2; the parity statement of Proposition 6.2; the start of the proof of Proposition 11.6; the density claim of Remark 18.5; the scope of Warning 14.3; and the crossover conjecture, withdrawn in Remark 15.14.

17. Strategy and the target theorem

Remark 17.1 (Ordered strategy).
(i) Difference, not commutator: build $(U_h - I)D$, split off the columns $d \mid h$ exactly by Lemma 8.2, and send the rest to adaptive Haar.
(ii) Approximate observability: work in $V/\ker D_\Lambda$, measure the optimal residual by pseudoinverse and the dual (10.3); do not attempt global identification, which Thm. 10.4 excludes.
(iii) Irregular tree: choose cuts by changes of divisor density, levels of M, centres and residue boundaries; exploit (9.3).
(iv) Local p-adic structure: couple Prop. 13.1 with the factors L_p of (S4), treating $p \nmid h$ as a universal factor and $p \mid h$ as diagonal defects.
(v) Excursions last: use them only for the reflection-odd component and as one family of stopping blocks, subject to (Q1).

Objective 17.2 (Adaptive boundary observability)**.** For fixed h, construct trees $\mathcal{T}_N$, multiscale subspaces $W_N \subset H_{\mathrm{sf}}$ and rows Λ_N adapted to CRT boundaries, not to the signs of μ, such that

$$\left\| P_{W_N}(A_h - L^{\mathsf{T}} B_{h,N} L) P_{W_N} \right\|_{\mathrm{op}} = o(1), \tag{17.1}$$

with $B_{h,N}$ the variational minimiser, and simultaneously

$$\left\| P_{W_N^{\perp}} \mu \right\|^2 = o(N). \tag{17.2}$$

Estimate (17.1) is to be proved by reducing all volumetric contributions to zero through Theorem 9.3 and controlling only the boundary cocycles; (17.2) must come from prefix, level and squarefree identities, not from probabilistic regularity. Together they would give $C_N(h) = o(N)$.

Objective 17.3 (Variant with central divisor rows)**.** With $F_{N,y} = \begin{bmatrix} V_y \\ C_N^{\mathrm{div}} \end{bmatrix}$, truncated divisor equations together with centre rows fixed independently of the orientations, prove

$$J_h^{\mathrm{off}} = F_{N,y}^{\mathsf{T}} B_{h,N} F_{N,y} + R_{h,N} \tag{17.3}$$

with $B_{h,N}(d,e) = 0$ unless $(d,e) \mid h$, prime-by-prime CRT structure, $\|B_{h,N}\| = N^{o(1)}$, and $\sup_{s\in\{\pm1\}^z} |s^{\mathsf{T}} R_{h,N} s| = o(N)$, no definition using s_μ or $C_N(h)$. Since $F_{N,y} s_\mu = 0$ by Theorem 11.7, this gives $s_\mu^{\mathsf{T}} J_h^{\mathrm{off}} s_\mu = o(N)$; combined with the independent intrablock objective $\sum_j \sum_{n,n+h\in E_j} \mu(n)\mu(n+h) = o(N)$ it gives $C_N(h) = o(N)$.

17.1. **Energy capture and the squeeze.**

Proposition 17.4 (Exact energy of a leaf projection)**.** *Let $W \subset H_{\mathrm{sf}}$ be the space of vectors constant on the leaves I of a partition of $\{1,\dots,m\}$ into blocks of consecutive indices, with $k_I = |I|$. Then*

$$\|P_{W^\perp}\mu\|^2 = \sum_I \Big(k_I - 1 - \frac{2}{k_I} \sum_{\substack{i<i' \\ i,i'\in I}} \mu(a_i)\mu(a_{i'}) \Big). \tag{17.4}$$

Each summand is nonnegative and vanishes if and only if μ is constant on the leaf.

Proof. $P_W\mu$ is the leaf average, so, using $\mu(a_i)^2 = 1$, $\|P_W\mu\|^2 = \sum_I k_I^{-1} \big(\sum_{i\in I} \mu(a_i)\big)^2 = \sum_I \big(1 + \frac{2}{k_I} \sum_{i<i'} \mu(a_i)\mu(a_{i'})\big)$; subtract from $\|\mu\|^2 = \sum_I k_I$. Nonnegativity and the equality case are Cauchy–Schwarz on each leaf. □

Remark 17.5. The span of the constant and the Haar functions of a truncated tree is exactly the space of vectors constant on the leaves of the truncation, so Proposition 17.4 covers every multiscale subspace admitted in Objective 17.2.

Corollary 17.6 (The squeeze)**.** *Condition* (17.2) *holds if and only if*

$$\sum_I (k_I - 1) - \sum_I \frac{2}{k_I} \sum_{\substack{i<i' \\ i,i'\in I}} \mu(a_i)\mu(a_{i'}) = o(m). \tag{17.5}$$

Consequently:

- *(i) singleton leaves contribute nothing, so* (17.2) *holds trivially when all but $o(m)$ indices lie in singleton leaves; but then W_N is essentially all of H_{sf} and* (17.1) *is the full observability problem, subject to Theorem 10.4;*
- *(ii) on leaves of size $k \geq 2$ the condition requires the within-leaf correlations of μ along consecutive squarefree integers to be maximal, $\sum_{i<i'} \mu(a_i)\mu(a_{i'}) = \binom{k}{2}$ up to o of the leaf mass: the opposite of the cancellation the objective is designed to establish.*

Proof. Immediate from (17.4). □

Remark 17.7 (Long leaves are excluded unconditionally). For a leaf with $k_I \to \infty$, $\sum_{i\in I} \mu(a_i)$ is the sum of μ over an integer interval of comparable length, since non-squarefree integers contribute zero. For partitions to which the short-interval estimate of [7] applies, in particular on average over the offset of a grid, these sums are $o(k_I)$ in mean square, hence $\|P_W\mu\|^2 = o(m)$ and (17.2) fails by the full amount. The input is used here only to refute.

Warning 17.8. Objective 17.2 is superseded as stated. Its capture condition cannot be met by a subspace built independently of the signs of μ except by taking essentially all of H_{sf}, and on leaves of size at least two it demands correlations of the wrong sign. In addition, the claimed implication silently requires a norm condition: writing $\mu = \mu_W + \mu_\perp$,

$$\mu_W^{\mathsf{T}} L^{\mathsf{T}} B L\, \mu_W = (L\mu_W)^{\mathsf{T}} B (L\mu_W), \qquad L\mu_W = L\mu - L\mu_\perp, \qquad L\mu = D_\Lambda \mu \in \{0, e_1\}, \tag{17.6}$$

because $(D\mu)(r) = \sum_{d|r} \mu(d) = \mathbf{1}[r = 1]$; this term is $o(N)$ only if $\|B_{h,N}\| \, \|L\|^2 \, \|\mu_\perp\|^2 = o(N)$. The Haar–CRT vanishing of Theorem 9.3 is unaffected, but it must be used without an L^2 capture condition.

Remark 17.9 (What survives). The engine common to both objectives is the annihilation $D\mu = e_1$ of (17.6). Objective 17.2 wrapped it in a capture condition that is self-defeating by Corollary 17.6. Objective 17.3 uses it directly, through $F_{N,y} s_\mu = 0$ in Theorem 11.7, and requires uniformity over the fibre rather than capture. It is therefore the live target, and the boundary-cocycle estimate of Theorem 9.3 should be recast as a tool for bounding the remainder $R_{h,N}$ there. Uniformity over all of V is excluded by Theorem 10.4; uniformity over the fibre $\{\pm 1\}^Z$ is not.

Warning 17.10. The ordinary correlation $C_N(h) = o(N)$ at fixed $h \neq 0$ is open; the logarithmically averaged version is known [16], building on [7]. Any inequality that yields $o(N)$ directly must therefore contain a genuinely new estimate, not a conjugation or a rewriting. Remark 8.1 and Warning 11.3 record two ways of failing this test.

18. Negative results and failed routes

Remark 18.1. The scalar quadratic energy at a single level carries no local information; the defect is the collapse to one number, not the weight.

Remark 18.2. Statistics built from M alone weight zeros by decaying factors in the ordinate and therefore mix heights; no operation on $\log N$ repairs this, since $\log N$ is conjugate to differences of ordinates, not to ordinates.

Remark 18.3. The excursion decomposition passes to continued fractions and hence to a Jacobi matrix [3], which is real symmetric and lands in the orthogonal class, as in Rem. 15.13.

Remark 18.4. A cell-complex description of M exists [2] but does not reach two-point cancellation: the complex is multiplicative while shifted correlations are additive, and no map relates the complex at n to that at $n + h$.

Remark 18.5. Any operator whose spectrum is the set of ordinates must reproduce $N(T) \sim (T/2\pi)\log(T/2\pi)$; the bounds (S1) place $\lambda_{\max}(D_N)$ at scale $N^{1/2+o(1)}$. This alone does not imply a semicircular density, and the earlier claim to that effect is withdrawn. What it does imply is that the first N eigenvalues of D_N cannot be $\gamma_1, \ldots, \gamma_N$, since $\gamma_N \sim 2\pi N/\log N$ exceeds $C_\varepsilon N^{1/2+\varepsilon}$.

Remark 18.6. A dyadic tree of midpoints of $|M|$ satisfies Chapman–Kolmogorov, hence is Markov in the scale variable; a constraint at a coarse level therefore gives no information at finer levels beyond the endpoint values, and the degrees of freedom double at each refinement. The product structure that makes the tree null computable is exactly the property that forbids constraint propagation. This closes the route of propagating a short-interval input down the tree. The same applies to the Euler factorisation in the multiplicative direction: both natural indexings factorise, so neither transports a constraint across scales.

Remark 18.7. Any hierarchy of bounds on the Möbius walk must respect $M(x) = \Omega_\pm(\sqrt{x})$ with explicit constants [12]; no ε in such a hierarchy is removable.

Remark 18.8. Splitting an interval into two halves keeps an $O(1)$ CRT error in each half; it does not reduce it. A left–right difference can cancel the main term when the lengths agree, but leaves a difference of boundary errors of size $O(1)$ with no sign and no uniform cancellation. For halves defined in squarefree time the set is no longer an arithmetic interval and the mask must be inserted, which makes the CRT factorisation less clean, not more.

Remark 18.9 (The irregular route, closed as a family). Every divisor-to-shift transfer that uses only $o(N)$ rows of D (Corollary 10.5), compression by prime valuations (Proposition 13.5), irregular Haar without an additional arithmetic estimate of the boundary cocycles (Proposition 9.5), or separable approximations of rank $o(N)$ (Remark 7.8) has macroscopic operator error on the squarefree subspace and cannot imply $C_N(h) = o(N)$. The quantitative progress available remains for logarithmic averages [16, 14].

Heuristic 18.10. The one door left open inside this family is a collective signed cancellation of the CRT boundary error,

$$\sum_{d,e} c_d c_e \, \epsilon_{d,e,h}, \tag{18.1}$$

for coefficients tied specifically to μ. Obtaining $o(N)$ for this form already requires control of the global parity of large primes, which is exactly the content that local factors, central rows and the squarefree mask do not contain.

19. Open questions

(Q1) Does $\widehat{G}$ inherit the tensor factorisation (2.1) from G? The excursion blocks are defined by the zero set of M and respect no multiplicative structure, so clean inheritance is not expected; partial inheritance restricted to the local factors L_p with small p would be the interesting outcome. Everything in Sections 15–17 depends on it.

(Q2) Prove a norm estimate for the boundary CRT cocycle operator of Theorem 9.3, restricted to adaptive Haar subspaces and after extracting the exact local p-adic modes of Prop. 13.1, respecting Warning 9.6. Following Warning 17.8, this estimate is now sought as a bound for $R_{h,N}$ in Objective 17.3, not as the first step of the superseded Objective 17.2.

(Q3) Sharpen Prop. 11.6: is the exceptional set actually bounded, or can $\log_2 N$ be attained?

(Q4) Test the prediction that $1/p(1)$ is the minimum of the divisor-resolved sine-kernel fraction.

(Q5) Determine the growth of $Z(N)$. Lemma 12.4 and every statement with an $O(hZ)$ error depend on $Z = o(N)$, which is not proved here.

(Q6) Close the non-attained-supremum case of Rem. 2.2.

(Q7) Decide whether the non-commutative deformation of Warning 14.3 escapes (S8) in the scaling window $\vartheta \asymp \vartheta_c(N)$.

Notebook History

Version 1: Three-constraint model and exact bridge second moment; reflection fibre and the canonical parameter-free null; divisor-defect Gram form pushed to orientation bits; blindness theorem; scale-resolved statistic with exact mean and variance; negative results.

Version 2: Abstract added; introduction restructured into what is proved, what is input, and the methodological point; standing results listed with status; sine-kernel section rewritten around the non-tautology criterion, with three recoveries of the sine-kernel coefficient from measured quantities; extremal-abscissa gap narrowed.

Version 3: Operator layer added. Three spaces and their information ledger; divisor–shift boundary lemma with interval mass bound 1; exact CRT pullback; algebraic Haar on arbitrary trees with exact vanishing of the volumetric CRT term and parent–children cocycle additivity; exact observability criterion with optimal constant, operator-norm dual and semidefinite formulation; no-go for sublinear divisor rows; extension of the blindness theorem to half-constant kernels, with the shift shown to escape it; central dichotomy lemma; triangular central divisor matrix with the logarithmic exceptional set, isolating the Möbius line; intrablock second-difference and aggregate identities, discrete coarea, and a counterexample showing the centre does not determine a lag; exact local tables

at each prime; collapse theorem for multiplicative diagonal weights and the resulting dichotomy; target objectives stated.

Version 4: Audit of the target objective. Conditioning of the two-point expansion on central data; redundancy of quarter and octave evaluations; exact leaf-energy identity for leaf projections in the squarefree space; the squeeze corollary showing that the capture condition of the adaptive boundary objective forces essentially the whole squarefree space or maximal positive within-leaf correlation; exclusion of long leaves by short-interval cancellation; the hidden norm condition in the claimed implication. The adaptive boundary objective is marked superseded; the central-row objective, built on $D\mu = e_1$ and uniformity over the fibre, is the live target.

Version 5: External audit incorporated after line-by-line verification. Pushed form corrected to $F^{\mathsf{T}}GF$; single orientation basis; conditioning on $M(N)$ made explicit; single-reflection parity corrected; exceptional-set proof started at $z_1 = 2$; semicircle claim and general weight dichotomy withdrawn; crossover conjecture withdrawn by the gauge obstruction, with a flux question and a crossover-scale heuristic in its place. New: the energy kernel as Montgomery's weight at $\alpha = 0$ with explicit density and tail; exact fourth-order relation between the Hermitian collision constant and the Möbius energy constant; $\mathcal{A}_0'$ as a discrete Lévy area, with its increment and the function $F(s)$; logarithmic energy as an aggregate of shifted correlations; compressed selection to at most $2N$ orientations and unique selection after $\lfloor \log_2 N \rfloor + 1$ further rows.

Version 6: Uncompressed irregular route executed. Boundary lemma restricted to the interior operator, with the boundary rows made explicit; maximal rank of the interior operator; exact $1 \to 1$, $\infty \to \infty$ and Hilbert–Schmidt norms; residue-boundary form of each column with the optimal variation bound; tree-independent Haar energy; quantitative observability no-go with constant $\mathfrak{S}_h/4$; signed local sums and the Mirsky expansion; failure of the shift to descend to valuations; maximal blind class as the additively separable kernels, with the distance of the shift and the Eckart–Young bound; global and block second-difference identities with explicit boundary terms. The irregular route is closed as a family; the collective CRT cancellation is recorded as the remaining heuristic.

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